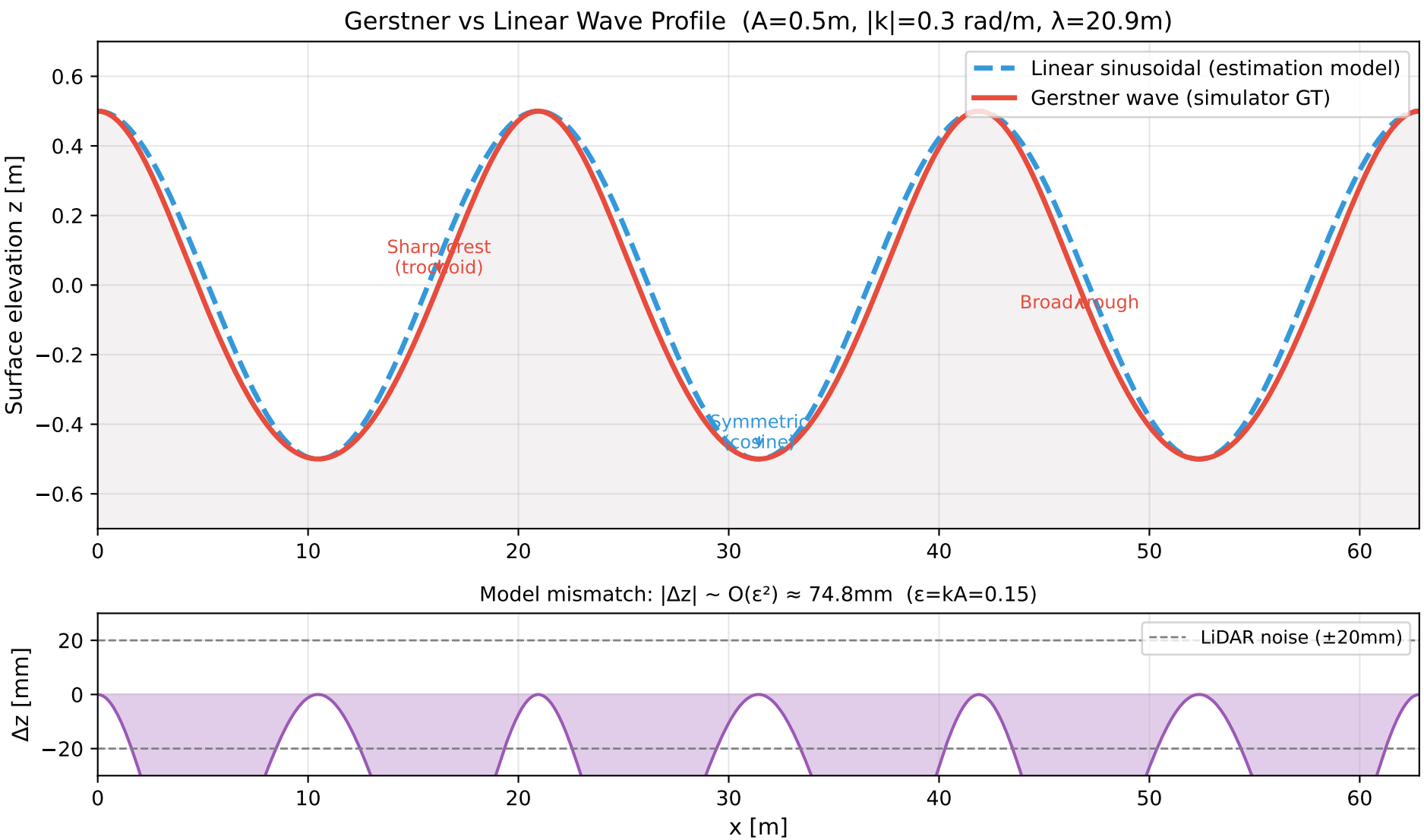

Gerstner vs Linear Wave Profile (A=0.5m, |k|=0.3 rad/m, λ=20.9m)
Linear sinusoidal (estimation model)
Gerstner wave (simulator GT)
Surface elevation z [m]
Sharp crest
(trochoid)
Broad trough
Symmetric
(cosine)
Model mismatch: |Δz| ~ O(ε²) ≈ 74.8mm (ε=kA=0.15)
LiDAR noise (±20mm)
Δz [mm]
x [m]